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資料結構

Data Structure

Lab 02

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Lab02-Ex1.
Count operations in Find Maximum code as comments.
Code

```
#include <iostream>
#include<vector>
using namespace std;

int stepCount = 0;

int findMax(const vector<int>& arr) {

  int max = arr[0];
  stepCount++//Assignments:1



  stepCount++;//loop initialization-i=1:1
  for (int i = 1; i < arr.size(); i++) {
    stepCount ++; // Loop comparison: i < arr.size():n
    stepCount ++; // Array access: arr[i]:n-1



    stepCount ++; // Comparison with target: arr[i] == target:n-1
   if (arr[i] > max) {
      max = arr[i];
      stepCount ++; //Assignments:n-1
    }
    stepCount ++; // Increment-i++:n-1
  }

  stepCount++;//return:1
  return max;
}

//time complexity is 5n-1
// Total operations:
//  5n-1 operations
// Therefore, O(n) complexity
```

Lab02-Ex2.
Integrate actual count variable in PrintPairs code.
Code

```
#include <iostream>
#include<iostream>
using namespace std;

int count = 0;

void printPairs(const vector<int>& arr) {
  count++//loop initialization-i=0:1
  for (int i = 0; i < arr.size(); i++) {
    count++//access array:n
    count++//Comparisons i < arr.size():n
    acount++ // inner loop initializationi=1:n
    for (int j = i + 1; j < arr.size(); j++) {
      count++//Comparisons j < arr.size():n(n+1)/2
      cout << arr[i] << "," << arr[j] << endl;
       count++//Increment-j++:n(n+1)/2
    }
  count++//Increment-i++:n
  }
}

//time complexity is n^2+4n+2
// Total operations:
//  n^2+4n+2 operations
// Therefore, O(n^2) complexity
```

Lab02-Ex3.

Count operations in (a) array copying code (b) array element counter code as comments.

Code

(a)

```
#include <iostream>
using namespace std;
int count = 0 ;
void copyArray(const vector<int>&
source, vector<int>& dest) {
    count++// loop initialization:1
  for (int i = 0; i < source.size();
i++) {
    count++//Loop comparison:n+1
    dest[i] = source[i];
    count++//assignment:n

    count ++; // Increment-i++:n
}
}

//time complexity is 3n+2
// Total operations:
//  2 + 3n operations
// Therefore, O(n) complexity
```

(b)

```
#include <iostream>
#include<vector>
using namespace std;

int stepCount = 0;

int countElement(const vector<int>&
arr, int target) {
  int count = 0;
  stepCount++;//assignments:1
  stepCount++;//loop initialization:1
  for (int i = 0; i < arr.size(); i++)
{
    stepCount++// Comparison : i <
arr:n+1
    stepCount++// Comparison with
target: arr[i] == target:n
    if (arr[i] == target) {
      count++;
      stepCount++//assignments:n
  }
    stepCount++// Increment-i++:n
  }
  stepCount++//return count
  return count;

}

//time complexity
// Total operations:
//  4 + 4n operations
// Therefore, O(n) complexity
```

Discussion Section
2.n 的最高次項在 n 極大時會主導整個 time complexity 的大小，此時常數幾乎可以忽略不計 2.1 拋棄係數及常數項與低次項，只看影響最大的因子，當 n 極大時，係數及低次項小到可以忽略不計 2.2 能不考慮硬體的前提下驗證演算法效能、減少不必要的縮減計算次數(優先降低影響最大因子) 3. 3.1 sumArray:5n+4、findNumber:5n+3，sumArray 比 findNumber 多 1 次運算 3.2 不會，他們的 Big-O 都是 O(n)